

Synopsis: Hysterical Response Halo and Galactic Scaling

What the theorems say (without the proofs)

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Computational note. Proofs, exploratory experiments, and paper writing were assisted by generative AI. Mathematical theorems stated in this paper are formally verified in Lean 4 in the companion specifications, as faithful ports of the TeX arguments at the abstractions recorded there, except results explicitly tagged as *literature hypotheses* (standard theorems cited from the literature and not re-proved here) or *physical inputs* (modeling assumptions outside mathematics).

Abstract

This is a short guide to the main claims of the galactic modelling paper. It builds on the Hysterical Response of the earlier response-theory paper. Each section says *why* the claim matters, then states the result. The proofs live in the full paper.

1 From Hysterical Response to a galactic halo

The earlier response-theory paper isolates the Hysterical Response: the residual force of an already-existing interaction under exponentially stable open-system dynamics,

$$F_H = -\text{Tr}(F \mathcal{L}^{-1} \mathcal{S}), \quad (1)$$

equivalently $\delta\rho_{ss} = -\mathcal{L}^{-1}\mathcal{S}$. No new force carrier is introduced: F_H is a component of the response of the ordinary force F itself. That paper stops at channel geometry, neutrality, instability, and infrared equivalence. It does *not* build galaxies.

The present paper takes that gravitational Hysterical Response as the force and continues it into an explicit spatial halo. A ballistic Wigner representation of the residual response (production source S , loss rate Γ) is phase-space bookkeeping for how the residue is organized in space—not a claim of a new messenger particle. Here S is the ballistic image of the companion paper’s residual drive $\mathcal{S}$. The retarded characteristic solution of $v \cdot \nabla f = S - \Gamma f$ is $f(x, v) = \int_0^\infty e^{-\Gamma t} S(x - tv, v) dt$ (Lean-checked under standard regularity). The total production rate $Q = \int J$ becomes the amplitude that later sets galactic scales. If that ballistic J packages Paper 1’s unentangled-incoming drive, then $Q = \alpha J_{\text{in}}^{\text{tot}}$; when baryonic accretion supplies those arrivals, $Q = \alpha \dot{N}_{\text{acc}}$ (conversion α remains model-dependent—not the unit identity $Q = \dot{M}_{\text{b}}$). “Hysterical Response Halo” means that extended residual-response density (and the acceleration it produces under the infrared closure)—ordinary gravity’s leftover, not a new force law invented here from scratch.

2 Two galactic problems

Observed rotation curves stay roughly flat where Newtonian baryons would already be Keplerian, and the asymptotic speed tracks baryonic mass as $M_b \propto v_f^4$ (BTFR). Those are different requirements: one is a radial profile $g_{\text{resp}} \propto 1/r$; the other is a mass normalization. The full paper separates them.

3 Ballistic response halo

Under an infrared ballistic realization—local velocity isotropy plus a single characteristic speed v —residual-response production takes the monoenergetic shell form $S = J \delta(|u| - v)/(4\pi v^2)$. For a localized, centered spatial profile J , the stationary density then has a universal leading far field.

Theorem 3.1 (Ballistic response halo). *Outside the source and well inside the propagation length,*

$$n = \frac{Q}{4\pi v r^2} + O(r^{-4}).$$

The leading halo is spherical; source shape enters only at higher order.

4 Why $g_{\text{resp}} = CQ/r$

A density $n \propto Q/r^2$ is not yet a force; the force is already the Hysterical Response of gravity. The primary modelling closure is the linear Poisson law for the associated response potential

$$\nabla^2 \Phi_{\text{resp}} = \kappa n. \quad (2)$$

Together with the ballistic halo and $\nabla^2 \ln r \propto r^{-2}$, this yields

$$g_{\text{resp}} = \frac{CQ}{r} \quad (3)$$

for a coupling $C = \kappa/(4\pi v) > 0$. Separate-field or unified-field language may be used as brief physical readings of that same infrared law; they do not independently postulate CQ/r .

Theorem 4.1 (Flat response rotation). *Circular balance $v_c^2/r = CQ/r$ then implies $v_c^2 = CQ$, hence an asymptotic response-dominated speed $v_f^2 = CQ$.*

Flat curves are therefore the radial consequence of the Hysterical Response halo plus the Poisson closure; they do not by themselves fix how Q scales with baryonic mass.

5 Capacity attractor vs formation conjecture

The ballistic spatial model fixes the shape of $n(r)$, not the pair (Q, M_b) . Introduce a capacity coefficient $A > 0$ and filling fraction $y = A^2 M_b / Q^2$ (equivalently the capacity inequality $M_b \leq Q^2 / A^2$ when $y \leq 1$). Response-limited assembly is modelled by a logistic ODE for y with stable fixed point $y_* \in (0, 1]$.

Corollary 5.1 (Square-root attractor). *Given that capacity inequality / filling setup and the stable assembly dynamics, at y_* one has*

$$Q^2 = \frac{A^2}{y_*} M_b, \quad (4)$$

so $Q \propto \sqrt{M_b}$. This attractor is a theorem/corollary of those hypotheses—not a conjecture.

Combined with $v_f^2 = CQ$,

$$v_f^4 = C^2 Q^2 \propto M_b. \quad (5)$$

Corollary 5.2 (BTFR exponent). *On that attractor, $v_f^4 \propto M_b$. The zero point remains encoded in C , A , and y_* ; those microscopic values are not derived here.*

Conjecture 1 (Formation / retention capacity). What is conjectural is the physical selection rule: galactic formation and mature retention require

$$Q \gtrsim O(\sqrt{M_b}).$$

That requirement is not proved from the ballistic spatial model (the attractor corollary places mature systems *on* $Q \sim \sqrt{M_b}$ once capacity filling is assumed).

6 Late-time size–mass law

With isotropic accretion ($j \propto M_b^{2/3}$) and size proxy $R \sim j/v_f$, the BTFR velocity scaling implies a definite size–mass exponent. The isotropic- j input is physically motivated by late-time feeding from a homogeneous intergalactic dust/gas reservoir (no preferred inflow direction); filamentary anisotropy would change the exponent.

Theorem 6.1 (Late-time size–mass). *On the mature attractor $v_f^4 = KM_b$,*

$$R \propto M_b^{5/12}.$$

Observed late-type slopes typically range from ~ 0.22 (CANDELS late-types) through ~ 0.39 (high-mass SDSS late-types) to ~ 0.385 (baryonic disc scale lengths). The prediction $5/12 \approx 0.42$ sits near the steeper end of that range.

7 Asymptotic oblateness

The first nonspherical correction for a planar axisymmetric source is a transport quadrupole. Under the same primary Poisson closure it produces an oblate potential whose effective flattening satisfies

$$q_{\text{eff}}^{-2}(R) = 1 + \frac{s}{R^2} + O(R^{-4}),$$

with $s = \langle R'^2 \rangle$ (and $s = 6R_d^2$ for an exponential disc). Thus the response is oblate at finite radius and becomes spherical as $R \rightarrow \infty$.

8 What is theorem vs assumption

- From the earlier Hysterical Response paper: $F_H = -\text{Tr}(F\mathcal{L}^{-1}\mathcal{S})$ of ordinary gravity; no new force carrier.
- IR ballistic shell: local velocity isotropy + single IR scale $\Rightarrow S = J\delta(|u| - v)/(4\pi v^2)$.
- Ballistic spatial model of that residue: r^{-2} halo and quadrupole (continuum asymptotics tagged where Lean does not re-prove the integral expansion).
- Modelling closure: primary Poisson $\nabla^2\Phi_{\text{resp}} = \kappa n \Rightarrow g_{\text{resp}} = CQ/r$ (and the same map for q_{eff}); separate/unified as readings only.

- Transport PDE: retarded characteristic solution solves $v \cdot \nabla f = S - \Gamma f$ (Lean).
- ODE / attractor (theorem): given capacity inequality and filling dynamics, stable y_* and $Q^2 = (A^2/y_*)M_b$, hence BTFR exponent.
- Accretion $\rightarrow Q$ (response-theory link): $Q = \alpha J_{\text{in}}^{\text{tot}} = \alpha \dot{N}_{\text{acc}}$ when accretion supplies unentangled arrivals; not the unit identity $Q = \dot{M}_b$.
- Conjecture (not from the spatial model): formation/retention requires $\boxed{Q \gtrsim O(\sqrt{M_b})}$.
- Physical inputs (angular momentum): isotropic $j \propto M_b^{2/3}$ (homogeneous intergalactic dust/gas) and $R \sim j/v_f$ give $a = 5/12$.
- Not claimed: new force carriers; microscopic C, A, α ; absolute BTFR zero point; Galactic numerical fits; proof of the formation requirement from the spatial model.